POISONED DATASET

Who changed the labels?

A tabletop game about data poisoning

D. Haydn Stucker · AIPI 590 · Explainable AI · Duke University · Fall 2026



1 The question

With four trusted checks, does auditing conflicting look-alike labels protect a learner better than random checks?

Poisoned labels can teach the wrong rule. When verification is costly, which labels should we check?

2 The game

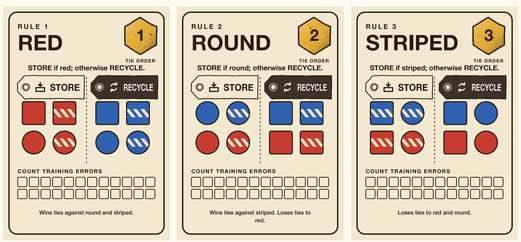
Two players swap roles. In each 24-card round, the attacker flips up to four STORE / RECYCLE tags.



Poison: flip up to 4 tags · **Inspect: count errors** · **Audit: up to 4 tokens** · **Lock: fewest errors** · **Deploy: 8 new cargo**

Audit, lock the rule, then test on eight held-out cards. Higher deployment score wins.

3 The learner



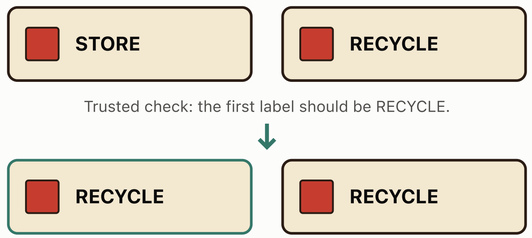
The robot picks the rule with the **fewest errors** against current labels. On eight held-out cards: true rule 8/8; wrong rule 4/8.

> **Explainability in play:** visible error counts explain how a corrected label can change the selected rule.

Related work: Vassilev et al., *Adversarial Machine Learning: A Taxonomy and Terminology*, NIST AI 100-2e2025. Jha, Hayase and Oh, *Label Poisoning is All You Need*, NeurIPS 2023.

4 What to audit?

Same features. Different labels.



Repair one label, then recount the rule errors.

- **Random:** uniformly check an unchecked card.
- **Conflict:** prioritize disagreement among identical cards, especially with verified labels. Recompute after each check; without conflict, choose randomly.

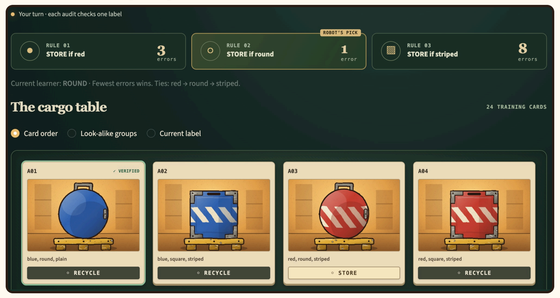
Threat model & experiment

Attacker: knows truth and learner; flips labels once before audits. **Defender:** sees features and labels; pays one token per trusted label.

Correlated decks link feature values; **independent** decks sample them separately. **Directed** attacks imitate a wrong rule; **random** attacks flip arbitrary labels.

200 paired seeds per condition; three policies on the same attacks (36,000 policy runs). Protocol v1.0 frozen before evaluation; 95% bootstrap intervals over seeds.

Print it or play on two devices



Online error counts update after audits. Also available as a reusable print edition.

5 Results

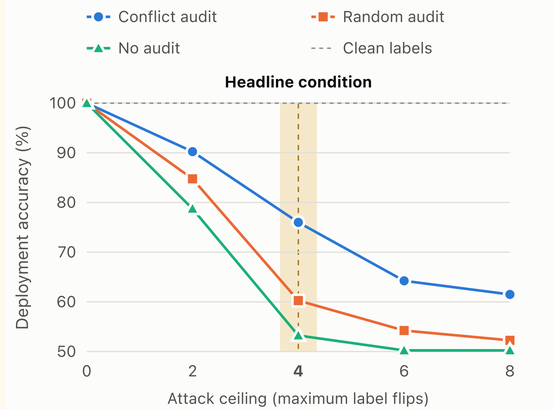
+15.8 percentage points of deployment accuracy

76.0%	60.3%
Conflict auditing	Random auditing

95% CI +12.0 to +19.5 percentage points.
Primary condition: correlated decks, directed attack, **up to 4 flips**, 4 audits, 200 seeds.

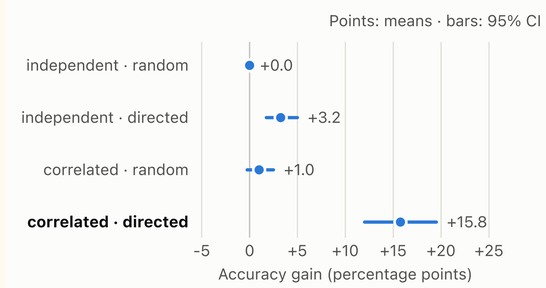
91 vs 28	73 · 117 · 10
true rules restored (conflict vs random)	seeds: conflict wins · ties · losses

As the attack budget grows



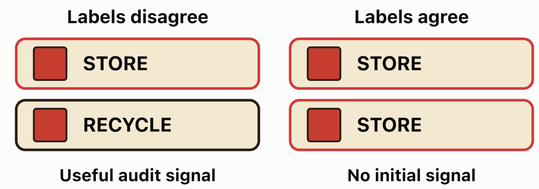
Four audits; shaded band marks the headline condition.

Conflict auditing's advantage



Up to four flips, four audits. Exploratory intervals crossing zero do not establish an advantage.

6 Where it fails



Red outlines: poisoned labels. True label: RECYCLE.

True-rule restoration (%) · 187 rules changed by attack

After attack	Seeds	Conflict	Random
Look-alikes disagree	78	87%	23%
No disagreement left	109	21%	9%

Exploratory split. A random find can reveal a new conflict.

Flipped by the attacker (true rule: round)

C23 STORE · C19 STORE · C12 STORE · C15 STORE

Conflict audit: learned red (wrong)

1. C11	2. C13	3. C02	4. C05
was right	was right	was right	was right

Random audit: learned round (restored)

1. C02	2. C21	3. C23	4. C12
was right	was right	found flip	found flip

Seed 1040: no initial conflict. Four clean audits missed the poison; random auditing found two flips.

7 What it means

Audit disagreement first. Agreement alone does not prove labels are correct.

- Three rules, 24 cards, a perfect oracle at uniform cost.
- The attacker ignores the defense; deliberately hiding conflicts could reduce the advantage.
- Simulated policies, not measured human play.

Game, rules & results

github.com/haydn-s/poisoned-dataset-game

Scan to get print files, setup instructions, raw results and reproducible figures.